

LECTURE 3.1



Adding Controls

ECONXXXXX Econometrics I

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PLACEHOLDER TERM

Where We Are

1. Foundations

1.1 Data and Variation

1.2 Conditional Means

1.3 Sampling Variation

2. Simple Regression

2.1 Fitting a Line

2.2 Goodness of Fit

2.3 Bias and Precision

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4.1 Diagnosing Violations

4.2 Limits of Regression

The First Lecture With Two Regressors

Lectures already given are greyed and today's is marked. Block 2 fitted one line four times over and reported it. Today puts a second variable into that line, and Lecture 3.2 changes what the variables themselves are.

What Lecture 2.4 Left Open

Lecture 2.4 finished with a defensible sentence about one number: a return of 1.5227 euro a year, a standard error of 0.1897, and an interval that covers the true 1.30.

Every word of that rests on the five assumptions of Lecture 2.3, and on the fourth in particular: at every level of schooling, the error averages to zero.

Today is the first lecture in which that assumption is false on purpose. Ability raises pay, it is not in the data, and people who stayed longer in school have more of it.

Question

If the thing you left out raises pay and also raises schooling, is your estimate of the return too large or too small? Answer before the arithmetic.

Lecture Plan

1 Partialling Out.

What a second variable does to a coefficient, written as a formula you can run.

2 Reading a Partial Coefficient.

What holding other things fixed covers, and three variables that break it three ways.

3 Omitted Variable Bias.

The gap between the regression you ran and the one you wanted, as a product of two slopes.

SECTION 1

Partialling Out

1. Partialling Out
2. Reading a Partial Coefficient
3. Omitted Variable Bias

The Two-Regressor Model for Earnings and Schooling

We want the return to a year of schooling among people of equal ability, so the population model carries two slopes rather than one:

$$y_i = \beta_0 + \beta_1(x_i - 13) + \beta_2 a_i + u_i \quad (1)$$

- y_i : hourly earnings of person i , in euro
- x_i : years of schooling of person i
- a_i : ability of person i , on the generator's own scale
- β_1 : the return to a year of schooling, holding a_i fixed
- β_2 : the return to a unit of ability, holding x_i fixed
- u_i : everything else that raises or lowers earnings

Assumption four now conditions on both regressors at once, $E[u \mid x, a] = 0$, and the five assumptions are renumbered MLR.1 to MLR.5 (Wooldridge, 2020).

Ability Is in the Data Only Because We Wrote the Population

No survey of Irish earnings carries a usable measure of ability, and the whole of section 3 is about what to do when it is missing. It is available today for one reason only: the generator produced it, so we can read it off.

That makes this section a controlled demonstration. We can run both regressions, put them side by side, and check by arithmetic what adding the second one did. Section 3 removes the column again and asks what survives.

A control is any regressor included so that the coefficient you care about can be read as a comparison at fixed values of it. Nothing about the arithmetic marks a variable as a control; that is a decision you make and defend.

A Simulation Is Not an Argument About Ireland

Every number today is checkable because the population was written rather than observed. That is what lets us verify a formula. It is not evidence about the Irish return to schooling, and no frame today claims it is.

The Part of Schooling That Ability Cannot Account For

A closed form for $\hat{\beta}_1$ in two regressors is not writable in scalars. What is writable is the variable the regression has left to use, so regress schooling on ability first:

$$x_i = \hat{\pi}_0 + \hat{\pi}_1 a_i + r_i \quad \sum_{i=1}^n r_i = 0 \quad \sum_{i=1}^n a_i r_i = 0 \quad (2)$$

- r_i : the part of person i 's schooling that ability leaves over
- $\hat{\pi}_1$: the slope from regressing schooling on ability

The two sums are the first-order conditions of that auxiliary fit, exactly the pair Lecture 2.1 wrote for earnings on schooling. They say that r_i carries no ability in it at all (Wooldridge, 2020).

Regression Anatomy

$\hat{\beta}_j$ is the slope from a simple regression of earnings on the part of x_j that the other regressors leave over.

The name and the phrase “other things equal” are both from Angrist & Pischke (2015), and it is the sentence to be able to say back.

The Two-Step Residual Recipe, Run on Our Own 120 People

Regression anatomy is not a description of what the machine is like. It is an identity, and it is computable in two lines:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n r_i y_i}{\sum_{i=1}^n r_i^2} \quad (3)$$

- $\sum r_i y_i$: the covariance of earnings with the leftover schooling
- $\sum r_i^2$: the variation in schooling that ability leaves over

These are the first numbers of the lecture, and each demonstration today is a separate run of the generator with one switch on, each with its own population of 200,000 and its own sample of 120. Numbers from two different runs are never the same regression under different controls.

Three Regressions, and the Last Two Agree Exactly

Earnings on schooling alone gives 1.5201. Earnings on schooling and ability together gives 1.1022.

Residualise schooling on ability, then regress earnings on that residual alone, and the slope is 1.1022.

Why the Two Steps Return the Same Coefficient

The agreement to four decimals is not a coincidence of this sample. Start from the long fit, which defines its own residuals, and sum:

$$\begin{aligned}\sum_{i=1}^n r_i y_i &= \sum_{i=1}^n r_i (\hat{\beta}_0 + \hat{\beta}_1(x_i - 13) + \hat{\beta}_2 a_i + \hat{u}_i) \\ &= \hat{\beta}_0 \sum_i r_i + \hat{\beta}_1 \sum_i r_i(x_i - 13) + \hat{\beta}_2 \sum_i a_i r_i + \sum_i r_i \hat{u}_i \\ &= \hat{\beta}_1 \sum_{i=1}^n r_i^2\end{aligned}\tag{4}$$

The first and third sums are the auxiliary conditions from the previous frame. The fourth is zero because r_i is built from schooling and ability, and the long fit's residuals are uncorrelated with both in the sample. The second is $\sum r_i^2$, because $\sum r_i x_i = \sum r_i (\hat{\pi}_0 + \hat{\pi}_1 a_i + r_i)$.

Raw and Residualised Scatterplots Side by Side

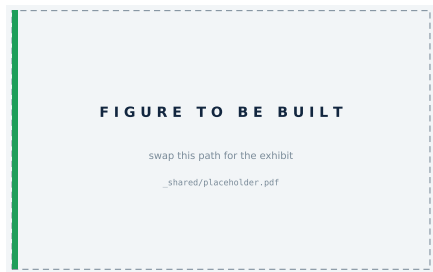


Figure 1: Earnings on raw schooling, 1.5201, and on the schooling that ability leaves over, 1.1022.

The Right-Hand Axis Is Narrower Than the Left-Hand One

Some of the variation in schooling has already been used by ability, so what is left to identify the slope is less than what we started with. That is the whole cost of a control, and the next frame prices it.

What a Control Costs in Precision

The denominator of the anatomy formula is also the denominator of the variance, which is why adding a variable and losing precision are the same event:

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum_{i=1}^n r_i^2} = \frac{\sigma^2}{S_{xx} (1 - R_1^2)} \quad (5)$$

- R_1^2 : the R-squared from regressing schooling on the other regressors
- S_{xx} : the sum of squared deviations of schooling, as in Lecture 2.3

Lecture 2.3's formula is the case $R_1^2 = 0$. Every control with any predictive power for schooling raises R_1^2 , shrinks the denominator and widens the sampling distribution (Wooldridge, 2020).

A Control Is Paid For in Standard Errors

Unbiasedness and precision are traded here, and the trade is not free in either direction. Lecture 3.3 gives the extreme case its name; today it is enough that the price is visible in the denominator rather than asserted beside it.

A Control That Almost Perfectly Predicts Schooling

Suppose ability predicts schooling almost exactly in our 120 people, so every r_i sits very close to zero. Nothing about earnings has changed.

Question

What happens to the standard error on the coefficient for schooling, and which part of the formula did you reason about to get there?

SECTION 2

Reading a Partial Coefficient

1. Partialling Out
2. Reading a Partial Coefficient
3. Omitted Variable Bias

What Holding Other Things Fixed Actually Covers

Nobody sampled 120 people of identical ability. The data arrived as a random sample, with ability taking whatever values it took:

$$\Delta \hat{y} = \hat{\beta}_1 \Delta x \quad \text{when} \quad \Delta a = 0 \quad (6)$$

- $\Delta \hat{y}$: the difference in fitted earnings between two people
- Δx : the difference in their schooling

Wooldridge's own words are that multiple regression lets us mimic a comparison at fixed ability without restricting the values of any regressor, and that the data were not collected in that fashion (Wooldridge, 2020).

Holding Fixed Reaches Exactly as Far as Your Columns

The comparison is between two people who differ in schooling and agree on ability. It says nothing about anything you did not measure, and it is arithmetic rather than evidence about what would happen if schooling rose.

Why Only One Coefficient in Your Table Is Interpretable

Every slope in a two-regressor table is a partial effect, and that is precisely why the table is not a list of comparable effects:

- $\hat{\beta}_1$ compares people who differ in **schooling** and agree on ability.
- $\hat{\beta}_2$ compares people who differ in **ability** and agree on schooling.

The set of things held fixed is different for each row, so the two numbers answer two different questions about two different comparisons. Reading down the column as though each row were an effect of its own variable is a category error.

Only one of them was designed. You put ability in to make assumption four credible for schooling. Nothing in the model was chosen to make it credible for ability, and schooling is a mediator for ability: one of the routes by which ability raises pay is that able people stay in school longer, and $\hat{\beta}_2$ has that route removed.

The Coefficient You Built the Model For Is the One You May Read

Report one coefficient causally and the rest as what they are, the adjustments that made it credible. A control's own coefficient almost never carries a defensible interpretation, and it is printed anyway.

Confounder, Mediator and Collider in One Causal Diagram

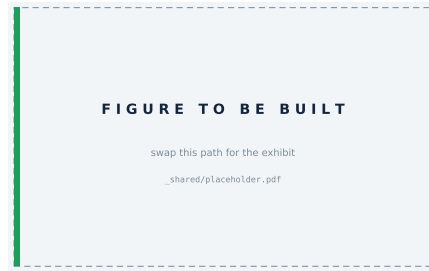


Figure 2: Ability causes both; occupation sits between them; promotion is caused by both.

One Command, Three Different Consequences

The same `lm()` call adds all three, and the regression cannot see which is which. Where a variable sits in the diagram is a claim about the world, and it has to be argued rather than tested.

Confounders: Controlling Removes a Bias

Able people stay in school for longer, and they also earn more for reasons that have nothing to do with school. So schooling and earnings rise together partly for a reason that is not the return to schooling.

A regression of earnings on schooling alone credits that reason to schooling. Putting the confounder in removes it: in the anatomy run, 1.5201 with ability left out and 1.1022 with ability included.

This is the case the phrase “adding controls” was invented for, and it is the only one of the three in which the coefficient ends up closer to the truth. Section 3 prices exactly this case, on a different run, once the column is taken away again.

A Confounder Causes Both, and Is Caused by Neither

That is the test, and it is a test about causal order rather than about correlation. A variable correlated with schooling and with earnings can equally well be a mediator or a collider, and those two make the answer worse.

Mediators: Controlling Removes Part of the Effect You Wanted

Schooling gets a person into a different occupation, and that occupation pays differently. This is not a nuisance to be removed. It is one of the routes by which schooling raises pay, and it is part of the number we said we wanted.

This is its own run of the generator, with the occupation switch on and its own sample. With occupation left out the coefficient is 2.5041. Control for occupation and it falls to 0.8936, so roughly two thirds of it is gone.

The missing two thirds did not disappear. It is the part of the return that travels through occupation, and the regression has handed it to the occupation coefficient instead.

The Smaller Number Is Not the Better Number

0.8936 is the return to schooling among people in the same occupation, which is a real quantity and is not the return to schooling. It deserves the name direct effect only if nothing unmeasured causes both occupation and pay; if something does, conditioning on occupation opens the problem on the next frame.

Colliders: Controlling Creates a Bias That Was Not There

Promotion is caused by schooling and it is caused by earnings. Nothing runs the other way. This is a third run of the generator, with the promotion switch on and its own population and sample.

With promotion left out, the coefficient on schooling is 1.1020, positive and close to the module's true return. Control for promotion and it is -0.9125 .

The sign reverses. Nothing was omitted from the second regression and nothing was mismeasured in either. The only difference between the two is the control, and the control is what produced the negative number.

This Is Why I Controlled for Everything I Had Is Not a Defence

Adding a variable made the answer worse, on a population we wrote ourselves. A defence has to name each control and say which of the three it is, and that claim cannot be read off the dataset.

Why the Collider Bias Is Not Sampling Noise

Promotion in this generator is a continuous index built from schooling and from earnings together. A person reaches a given value of it either by having stayed in school or by earning well.

So among people who share a value of the index, a person with less schooling must have arrived there the other way: by earning more. Holding the index fixed forces schooling and earnings to trade off against each other, and the regression reads that trade-off as a negative return.

Both regressions were run on the same rows, so the distance between 1.1020 and -0.9125 is not a difference between two samples at all. It is what conditioning on a common consequence did.

Conditioning Can Manufacture a Relation That Nothing Caused

No arrow in the diagram runs from schooling to earnings through promotion. The association appears only inside the regression that holds promotion fixed, and it would appear in a sample of any size (Wooldridge, 2020).

What Controlling Does in Each of the Three Cases

Three variables, one command, three results, and only the first is the one the phrase “adding controls” promises:

- **Confounder.** Controlling **removes a bias**: the part of the association that came from the common cause is taken out.
- **Mediator.** Controlling **removes part of the effect you wanted**: the route through the mediator is taken out with it.
- **Collider.** Controlling **creates a bias where there was none**: conditioning on a common consequence induces an association.

Two of the Three Are Made Worse by the Control

The regression output looks identical in all three cases: a coefficient, a standard error, a star. Which case you are in is settled by causal order, which is an argument you make before you fit and cannot recover afterwards.

Whether Promotion Belongs in the Regression

I want the return to schooling. The dataset has an index of whether the person was promoted in the last three years, and people high on it both earn more and have more schooling.

Question

Should it go into the regression, and what would you need to know about the world, rather than about the data, to settle it?

SECTION 3

Omitted Variable Bias

1. Partialling Out
2. Reading a Partial Coefficient
3. Omitted Variable Bias

The Regression We Ran and the Regression We Wanted

Section 1 had ability because we wrote it. Take the column away, which is the situation of every real dataset, and the question is how far that puts the coefficient we can compute from the one we want:

$$\text{long: } y_i = \hat{\beta}_0 + \hat{\beta}_1(x_i - 13) + \hat{\gamma} a_i + \hat{u}_i \qquad \text{short: } y_i = \tilde{\beta}_0 + \tilde{\beta}_1(x_i - 13) + \tilde{u}_i \quad (7)$$

- $\hat{\gamma}$: the coefficient on ability in the long regression
- $\tilde{\beta}_1$: the slope on schooling from the regression that left ability out

Omitting a variable correlated with schooling is what makes assumption four fail, and this is the case Wooldridge derives rather than asserts (Wooldridge, 2020).

Hat, and Now Tilde

A hat marks an estimate, as it has since Lecture 1.3. A tilde marks an estimate from a regression that left something out. Both are estimates; only one of them is estimating the quantity we said we wanted.

Deriving the Bias in Sample Algebra

The two regressions are run on the same 120 rows, so the gap between them is arithmetic rather than a statistical claim. Substitute the long fit into the short formula:

$$\begin{aligned}\tilde{\beta}_1 &= \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) y_i = \frac{1}{S_{xx}} \sum_{i=1}^n (x_i - \bar{x}) (\hat{\beta}_0 + \hat{\beta}_1 (x_i - 13) + \hat{\gamma} a_i + \hat{u}_i) \\ &= \hat{\beta}_0 \underbrace{\frac{\sum_i (x_i - \bar{x})}{S_{xx}}}_{=0} + \hat{\beta}_1 \underbrace{\frac{\sum_i (x_i - \bar{x})(x_i - 13)}{S_{xx}}}_{=1} + \hat{\gamma} \underbrace{\frac{\sum_i (x_i - \bar{x}) a_i}{S_{xx}}}_{=\hat{\delta}} + \underbrace{\frac{\sum_i (x_i - \bar{x}) \hat{u}_i}{S_{xx}}}_{=0} \quad (8) \\ &= \hat{\beta}_1 + \hat{\gamma} \hat{\delta}\end{aligned}$$

The first two braces are Lecture 2.3's two facts about deviations of schooling. The fourth is the long fit's first-order condition for schooling. The third is the slope from regressing ability on schooling (Wooldridge, 2020).

The Auxiliary Slope Points the Other Way

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0 regresses ability on schooling. Section 1's $\hat{\pi}_1$ regressed schooling on ability. Two auxiliary regressions, 22

The Bias Is a Product of Exactly Two Slopes

Rearranged, the derivation says the short regression reports the long coefficient plus one product, and names both of its ingredients:

$$\underbrace{\tilde{\beta}_1 - \hat{\beta}_1}_{\text{bias}} = \hat{\gamma} \times \hat{\delta} \quad \hat{\gamma} : \text{ability to pay} \quad \hat{\delta} : \text{schooling to ability} \quad (9)$$

A product, so a zero in either ingredient gives no bias at all. A variable that raises pay but is unrelated to schooling costs nothing. Neither does one that predicts schooling well and has no return of its own. Both are needed.

The sign follows the two signs. Both positive, or both negative, and the short slope is too large. One of each, and it is too small (Wooldridge, 2020).

The Two Ingredients Are Arguable Separately

Each is a claim someone can dispute on its own terms: does the missing thing pay, and do the people who have it stay in school longer. That is a more useful argument than a dispute about whether the estimate is credible.

The Ability Population Is Not Block 2's Population

Block 2 ran the generator with every switch off, and its sample of 120 gave 1.5227, a standard error of 0.1897 and an R-squared of 0.3531, against a true return of 1.30.

The arithmetic on the next frame comes from a run with the ability switch on: its own population, its own sample of 120. The short regression there gives 1.6700, with a standard error of 0.1652, a t-statistic of 10.11 and an R-squared of 0.4642.

1.6700 Is Not What Happens When You Drop a Column From Block 2

Two different populations and two different samples, so 1.5227 and 1.6700 are not a before and an after. The same warning applies to section 1's 1.5201 and to the two bad-control runs: four separate worlds, four separate draws. The only pair on this deck that is a before and an after is the next frame's.

The Formula Closing on Our Own 120 People

The short and the long regression here are run on the same 120 rows, so the formula is checkable to the last digit rather than true on average:

$$\underbrace{1.6700}_{\tilde{\beta}_1} - \underbrace{1.2538}_{\hat{\beta}_1} = 0.4162 = \underbrace{0.2206}_{\hat{\delta}} \times \underbrace{1.8867}_{\hat{\gamma}} \quad (10)$$

A year more schooling goes with 0.2206 more ability, and a unit more ability is worth 1.8867 euro an hour. Their product is 0.4162, which is the whole distance between the two regressions, to four decimal places.

Nothing was fitted to make that work. The two ingredients come from two separate regressions that never saw each other, and the identity is what forces them to agree (Wooldridge, 2020).

The Formula Is an Identity, Not an Approximation

It holds in every sample, including ours, for any omitted variable and any sample size. What is unavailable in real work is not the formula: it is $\hat{\gamma}$ and $\hat{\delta}$, because computing either needs the column we do not have.

The Short and the Long Slope as Ability Is Made to Matter More

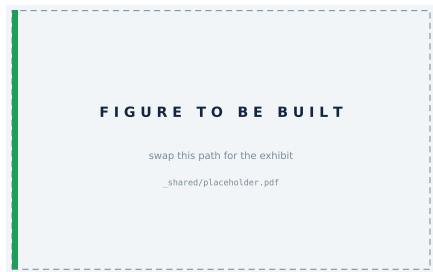


Figure 3: The short slope rises with the return to ability; the long slope stays where it is.

Only One of the Two Lines Rises

The long regression holds ability fixed, so the return to ability is irrelevant to it. The short regression has nowhere to put ability except the schooling coefficient, and the steeper the return to ability, the more of it lands there.

Any Pair of Slopes With the Same Product

Suppose someone rejects our estimate and claims the true return is one euro. That claim has a price: the bias would have to be 0.6700, and the objector owes us a pair of slopes whose product is 0.6700.

Many pairs do it, and that is the point rather than a weakness. The formula does not settle an objection; it converts it into two claims about the world that can be argued separately and, sometimes, checked against other evidence.

In practice the second ingredient is the harder one to defend. Almost everyone will agree that ability is worth something in pay. Arguing that it is worth three euro an hour per unit, on a scale nobody has measured, is a different matter.

A Bias of 0.6700 Requires Both of These at Once

A return to ability of 1.8867 needs an auxiliary slope of 0.3551 rather than 0.2206, which is over half again as much sorting into school by ability. Holding the sorting fixed instead needs a return to ability of 3.0372.

What Would Have To Be True?

The claim on the table is that 1.2538 is the return to schooling among people of equal ability, and that nothing else left out of the model does what ability does.

Question

What would have to be true for this number to mean what we just said it means? And what is the plausible story where it isn't?

Application

The app carries two sliders: how much the omitted variable raises schooling, and how much it raises pay. The short and the long coefficient are printed side by side as the sliders are set.

Question

With both sliders at their midpoint, is the short estimate above the long one, below it, or the same? Then set either slider to zero and answer again.

The lab for this lecture is the residual recipe: `resid()`, then `lm()`, checked against the two-variable `lm()`, and then the same check for the bias formula.

Summary: Partialling Out, Bad Controls and Omitted Variable Bias

1 Partialling Out.

A coefficient is the simple-regression slope on the part of a variable the other regressors leave over, which on our sample is 1.1022 either way.

2 Reading a Partial Coefficient.

Each row holds a different set of things fixed, and a control can remove a bias, remove part of the effect, or reverse the sign.

3 Omitted Variable Bias.

Short minus long is $\hat{\gamma}\hat{\delta}$, which on our 120 people is $0.2206 \times 1.8867 = 0.4162$.

Next Lecture

Lecture 3.2 keeps today's machinery unchanged and replaces the variables that go into it:

- **Logs and Elasticities:** change the units and the coefficient answers a different question.
- **Dummy Variables:** a category shifts the intercept, relative to a base you chose.
- **Interactions:** let the slope differ across groups, and every coefficient around it is read differently.

Regression anatomy carries over intact. A dummy variable is residualised in exactly the way ability was today, and the interpretation of an interaction is read off the same partial-effect rule.

Before Lecture 3.2

1 **Play:**

The app. Set the second slider to zero and see whether you can make the bias come back by any setting of the first.

2 **Apply:**

Problem set 3, covering 3.1 and 3.2. One paragraph defending a control you chose, one rejecting a control you did not.

3 **Review:**

Wooldridge (2020), sections 3-2 and 3-3, on the mechanics, the partialling-out interpretation and the bias.

Thank you

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BACKUP

Appendix

Derivation: The First-Order Conditions for Two Regressors

The anatomy proof used three conditions on the long fit's residuals without obtaining them. All three come from differentiating the sum of squared residuals and setting each derivative to zero:

$$\sum_{i=1}^n \hat{u}_i = 0 \quad \sum_{i=1}^n (x_i - 13) \hat{u}_i = 0 \quad \sum_{i=1}^n a_i \hat{u}_i = 0 \quad (11)$$

One Condition per Coefficient

Three coefficients give three conditions, which is why the divisor for $\hat{\sigma}^2$ is $n - 3$ here and was $n - 2$ in Lecture 2.3. Wooldridge writes the general set for k regressors as $k + 1$ equations (Wooldridge, 2020).

Derivation: Why the Centring Constant Cancels

Both body derivations used $\sum (x_i - \bar{x}) = 0$ and $\sum (x_i - \bar{x})(x_i - 13) = S_{xx}$, proved in the appendix to Lecture 2.3:

$$\sum_{i=1}^n (x_i - \bar{x})(x_i - 13) = \sum_{i=1}^n (x_i - \bar{x})^2 + (\bar{x} - 13) \sum_{i=1}^n (x_i - \bar{x}) = S_{xx} \quad (12)$$

Centring at the Leaving Certificate Costs Nothing

The second sum is zero, so the 13 contributes nothing and both derivations run unchanged on raw schooling. Centring sets what the intercept means and leaves every slope on this deck where it was.

Notes and Tools

- **The App:** two sliders, ability to schooling and ability to pay, with the short and long coefficients printed.
- **The Lecture in One Line:** `coef(lm(y ~ x + a))[2]` beside `coef(lm(y ~ x))[2]`.
- **This Lecture's Lab:** `resid()` and `lm()`, checked against the two-variable fit and against $\hat{\gamma}\hat{\delta}$.

References

Angrist, J. D., & Pischke, J.-S. (2015). *Mastering 'metrics: The path from cause to effect*. Princeton University Press.

Wooldridge, J. M. (2020). *Introductory econometrics: A modern approach* (7th ed.). Cengage Learning.